

# Practicum 3: Activation Functions

## Graphs, Domains, Ranges and Points of Non-Smoothness

### Objective

Implement the standard activation functions of neural networks from scratch, plot each function together with its derivative, and document analytically the domain, the range, every point of discontinuity and every point at which the function fails to be differentiable. The purpose is to make precise, before backpropagation is introduced, exactly where each activation can and cannot supply a gradient, and what convention must be adopted where it cannot.

### The Functions

Implement each of the following as a NumPy function of a real argument (or a real vector, for the softmax). Do not call library implementations; write the formula.

- (1) **Heaviside step:**  $H(x) = 1$  if  $x \geq 0$ , and 0 otherwise.
- (2) **Piecewise linear (saturating linear):**  $f(x) = 0$  for  $x < -1$ ;  $f(x) = \frac{1}{2}(x + 1)$  for  $-1 \leq x \leq 1$ ;  $f(x) = 1$  for  $x > 1$ .
- (3) **Sigmoid:**  $\sigma(x) = 1/(1 + e^{-x})$ .
- (4) **Hyperbolic tangent:**  $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$ .
- (5) **ReLU:**  $\text{ReLU}(x) = \max(0, x)$ .
- (6) **Rectified (clipped) ReLU, ReLU6:**  $\text{ReLU6}(x) = \min(\max(0, x), 6)$ .
- (7) **Softmax:** for  $z \in \mathbb{R}^K$ ,  $\text{softmax}(z)_k = e^{z_k} / \sum_{j=1}^K e^{z_j}$ .
- (8) **Leaky ReLU:**  $\text{LReLU}_a(x) = x$  for  $x \geq 0$  and  $ax$  for  $x < 0$ , with slope  $a = 0.1$  (and see Task 2 for  $a = 0.01$ ).

Plot every scalar function on the interval  $[-8, 8]$  with at least 2000 evenly spaced abscissae. The softmax is treated separately in Task 3.

### Assignment Tasks

#### Task 1: Function and Derivative Plots

For each scalar function (1)–(6) and (8):

- Derive the derivative  $f'(x)$  by hand, on every open interval on which the function is differentiable, and state the one-sided derivatives  $f'(x_0^-)$  and  $f'(x_0^+)$  at every point  $x_0$  where the two-sided derivative does not exist.
- Produce one figure with two vertically stacked panels sharing the horizontal axis: the function  $f$  in the upper panel and its derivative  $f'$  in the lower panel.
- Mark every point of discontinuity of  $f$  with an open circle at the excluded value and a filled circle at the attained value, so that the reader can see which one-sided limit the function takes.

- Mark every point at which  $f$  is continuous but not differentiable with a distinct marker (e.g. a filled square) on the upper panel, and show the corresponding jump in  $f'$  on the lower panel with open and filled circles as above.
- Draw the range of  $f$  as a shaded horizontal band on the upper panel, and annotate the band with the interval in standard notation, distinguishing open from closed endpoints (e.g.  $(0, 1)$  for the sigmoid,  $[0, 1]$  for the piecewise linear function).

## Task 2: The Property Table

Construct a single table with one row per function (1)–(8) and the following columns:

- Domain.
- Range, in interval or set notation, with correct endpoint brackets. (The range of the softmax must be described as a subset of  $\mathbb{R}^K$ ; be precise about whether the boundary of the simplex is attained.)
- Points of discontinuity, listed explicitly, with the left limit, right limit and function value at each.
- Points of non-differentiability, listed explicitly, with the left and right derivatives at each.
- The derivative on each maximal open interval of differentiability, as a formula.
- Whether the function is bounded, monotone (strictly or weakly), and zero-centred (i.e. whether  $f(0) = 0$  and  $f$  is odd).
- The set on which  $|f'(x)| < 10^{-3}$  (“saturation region”), computed exactly where possible and numerically otherwise.

Add one further row for Leaky ReLU with  $\alpha = 0.01$  and explain in a sentence why the kink at 0 is visually invisible at that slope although the non-differentiability is unchanged.

## Task 3: The Softmax

The softmax is a vector-valued map  $\mathbb{R}^K \rightarrow \mathbb{R}^K$ , not a pointwise function, so its graph cannot be drawn on a single pair of axes. Produce the following instead.

- $K = 2$ . Show algebraically that  $\text{softmax}(z_1, z_2)_1 = \sigma(z_1 - z_2)$ . Plot  $\text{softmax}(z)_1$  and  $\text{softmax}(z)_2$  against  $t = z_1 - z_2 \in [-8, 8]$  on one set of axes, and overlay the sigmoid from Task 1 to confirm the identity visually.
- $K = 3$ , **one-dimensional slice**. Fix  $z = (t, 0, 0)$  and plot the three components  $\text{softmax}(z)_1$ ,  $\text{softmax}(z)_2$  and  $\text{softmax}(z)_3$  against  $t \in [-8, 8]$ . Verify on the plot that the three curves sum to 1 at every  $t$ .
- $K = 3$ , **two-dimensional slice**. Fix  $z_3 = 0$  and produce a filled contour plot of  $\text{softmax}(z_1, z_2, 0)_1$  over  $(z_1, z_2) \in [-6, 6]^2$ , with the level sets  $0.1, 0.2, \dots, 0.9$  labelled.
- Shift invariance**. Prove that  $\text{softmax}(z + c\mathbf{1}) = \text{softmax}(z)$  for every scalar  $c$ , where  $\mathbf{1}$  is the all-ones vector. Use this to explain why (b) and (c) lose no generality by fixing one coordinate to 0.
- Range**. Prove that  $\text{softmax}(z)$  lies in the open simplex  $\{p \in \mathbb{R}^K : p_k > 0 \text{ for all } k, \sum_k p_k = 1\}$  for every finite  $z$ , and that no vertex of the simplex (a one-hot vector) is ever attained. Relate this to the range  $(0, 1)$  of the sigmoid.
- Jacobian**. Derive  $\partial \text{softmax}(z)_i / \partial z_j = \text{softmax}(z)_i (\delta_{ij} - \text{softmax}(z)_j)$ , and state whether the softmax has any point of discontinuity or non-differentiability in  $\mathbb{R}^K$ .

#### Task 4: Numerical Verification of the Derivatives

For each scalar function, compare your hand-derived  $f'$  with the central difference  $(f(x+h) - f(x-h))/(2h)$ ,  $h = 10^{-5}$ , at  $x \in \{-3, -1, -0.5, 0, 0.5, 1, 3, 7\}$ .

- Tabulate the analytic derivative, the central difference, and their absolute difference at every point.
- At each point of non-differentiability, the central difference returns a single finite number even though  $f'$  does not exist. Compute that number, show that it equals the average of the two one-sided derivatives, and explain why this is a coincidence of the symmetric stencil rather than a definition of the derivative.
- At the discontinuity of the Heaviside step, compute the central difference and explain why it blows up as  $h \rightarrow 0$ . Explain in two or three sentences why this makes the step function unusable as a hidden-unit activation in gradient-based training, even though it was perfectly usable in Practicum 1.

#### Task 5: Conventions at the Kink

For ReLU, ReLU6 and Leaky ReLU, the derivative at  $x = 0$  is undefined but a value must be returned by any implementation.

- State the set of all valid subgradients of each function at 0 (the subdifferential), as a closed interval.
- State the value actually returned at 0 by `torch.nn.functional.relu` and `jax.nn.relu` (look this up and cite the documentation or source), and confirm it lies in the subdifferential.
- Explain, in a short paragraph, why the choice made at a single point does not matter in practice for training with floating-point inputs, and describe one situation in which it would matter.

#### Required Deliverables

1. The seven two-panel figures from Task 1, with every discontinuity, non-differentiable point and range band marked as specified, each with a caption stating the domain and range in interval notation.
2. The property table from Task 2.
3. The four softmax figures from Task 3 (a)–(c) and the four proofs from Task 3 (d)–(f), typeset in full.
4. The numerical verification tables and explanations from Task 4.
5. The written answers to Task 5.

#### Submission Guidelines

- Implementation: Python with NumPy and Matplotlib only. Every activation function and every derivative must be written by hand; calls to library activation functions are permitted only in Task 5, for the purpose of reading off the value at 0. Your code must be included as an appendix and must regenerate every figure and table from a single run.
- Document Format: Compile all deliverables (figures, tables, derivations and proofs) into a single, well-organised PDF document.
- File Naming Convention: Name your final submission file strictly as `RollXX_FirstName_Assign3.pdf`, where XX is your class roll number; e.g. `Roll01_Adam_Assign3.pdf` for a person named Adam Smith having class roll 01.